



MIA Registration Project

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Group 23

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1 | Introduction

Medical image registration is an important component in medical image analysis, especially when aligning different image modalities such as T1-weighted and T2-FLAIR MR images. Intensity-based registration methods are commonly used for intermodal registration, since it provides more information and requires less user interaction compared to point-based registration methods. Mutual information (MI) is an effective similarity measure due to its robustness to intensity differences between modalities [3]. However, MI is sensitive to initialization and requires optimization [2]. Gradient-based optimization methods, such as gradient ascent, are commonly used for this. Their performance depends on the choice of parameters, such as the learning rate.

A fixed learning rate may lead to slow convergence or instability: if chosen too small, the maximum or minimum may not be reached in time, whereas a too large learning rate may lead to missing the maximum [1]. In contrast, varying the learning rate during the optimization method may provide a solution for this issue. The goal of this study is to investigate how the choice of learning rate strategy, fixed or varied, affects the performance of intermodal intensity-based image registration.

1.1 | Research question

What is the effect of a varying learning rate in the gradient ascent algorithm instead of a fixed one, on the performance of intermodal intensity-based registration, using mutual information as a similarity measure?

1.2 | Hypothesis

The hypothesis states that an adaptive learning rate will achieve equal or higher final MI compared to a fixed learning rate.

2 | Study design

This study investigates the effect of using a varying learning rate in the gradient ascent algorithm, compared to a fixed learning rate — on the performance of intensity-based intermodal image registration, using mutual information (MI) as the similarity measure. The registration is performed between the T1 and T2 MRI images.

We use the dataset from the MRBrainS medical image analysis challenge, which provides MRI brain scan slices in both T1 and T2 modalities. The dataset includes 3 patients, each contributing 3 transverse slices per modality, resulting in 9 T1–T2 slice pairs used for intermodal registration. This does provide a limitations since 9 slice pairs is a small sample size.

Our method uses affine registration, which accounts for rotation, translation, scaling, and shearing, making it appropriate for multimodal brain MR images. We optimise the affine transformation parameters using a gradient ascent algorithm. MI is chosen as the similarity metric because of its robustness in aligning images of different modalities.

We compare two versions of the gradient ascent algorithm: one with a fixed learning rate and another where the learning rate adapts during optimization. This choice stems from the hypothesis that an adaptive learning rate may lead to improved registration quality. To evaluate, the maximum MI will be compared between the two versions of the algorithm since a higher final MI would suggest a better alignment.

For the fixed learning rate a value of 0.001 was chosen as this was used in the course. Through trial and error this learning rate was found to be sufficient. The thresholds for the adaptive LR were chosen through trial and error. These thresholds were used to determine when the LR should be increased or decreased based on how close the MI is to its maximum, or if the MI is in a local optimum. The values of the thresholds will be further explained in the Methods section. The number of iterations was set to 150 to ensure convergence without being computationally wasteful.

3 | Methods

3.1 | Similarity index

In intensity-based image registration between multimodal images, mutual information (MI) is widely used as a similarity measure due to its ability to align images with different intensity distributions, which is crucial for inter-modal registration. MI measures the amount of information shared between two images A & B (the transformed image). This gives a better view of the registration compared to normalised cross-correlation. MI is calculated as:

$$MI(A, B) = \sum_{a=1}^n \sum_{b=1}^n p_{A,B}(a, b) \log \frac{p_{A,B}(a, b)}{p_A(a)p_B(b)} \quad (3.1)$$

The MI is calculated in *registration.py* by the following:

```
292 MI = np.sum(p * np.log2(p / (p-I @ p-J)))
```

3.2 | Code explanation

Important parts of code of *registration_project.py* to highlight. The application of the adaptive learning rate is mostly focused on the function *varying_learning_rate*

```
159 if improvement > 0 and abs(improvement - previous) > 0.02:
160     mu_out = max(mu_in * 0.95, min_mu)
```

The code above decides when to decrease the learning rate (LR). This happens when the MI is increasing, and the change in improvement compared to the previous step is large (> 0.02). This is because this suggests we might be overshooting the optimum. By reducing the learning rate, smaller steps are taken allowing for more precise convergence near the peak of the objective.

```
163 elif improvement < -0.05 and (improvement - previous) < -0.05:
164     mu_out = min(mu_in * 1.05, mu_0)
```

Next code increases the LR when the MI drops significantly (> 0.05) and the drop is worse than the previous step, because this suggest the gradient descent is on a bad path or maybe stuck in a local minimum. a larger learning rate helps make a bigger move and escape from this undesirable region.

```
168 elif MI_current < MI_local and abs(improvement) < 0.001 and abs(improvement - previous) <
    0.001:
169     mu_out = min(mu_in * 1.1, mu_0)
```

This code allows for an escape of a local optimum. If the MI is low (below the threshold of 0.8) and there has been no change in MI for 2 steps, we increase the learning rate by 10%. A low MI combined with stagnation suggest we're stuck in a local optimum. A larger step can jump over this local optimum and help move toward a better solution.

```
172 if abs(improvement) < 0.001:
173     plateau_counter += 1
174     if plateau_counter > 5:
175         mu_out = max(mu_in * 0.9, min_mu)
176 else:
177     plateau_counter = 0
```

By counting up the `plateau_counter` it's possible to check for little no improvement and slowly reduce the LR by 10%. A plateau indicates a flat region with low gradient. Reducing the LR can help detect subtle gradients and continue making progress.

By not only looking at the latest MI change (*improvement*), but also comparing it to the one before (*previous*) it creates a more robust mechanism of dynamically increasing or decreasing the learning rate.

4 | Experiments

To investigate the effect of a varying LR contrary to a constant LR, a data set retrieved from the MRBrainS medical image analysis challenge was used. The data consists of 18 traverse slices of MRI brain scans with two different sequences: T1-weighted and T2-FLAIR (3 patients $\times$ 3 slices per patient $\times$ 2 modalities). The T1 and T2 images formed sliced pairs that were then used for intermodal registration.

Using these pairs, 2 methods were performed on every image of each patient. First, a varying learning rate is applied, using the method as described below, and after that a constant learning rate is applied.

The varying learning rate was found by increasing and decreasing this value and noting changes to the MI. When the MI increased rapidly, the LR was reduced with a small amount in order to avoid overshooting. The LR was set to increase when the MI dropped significantly to avoid stagnation due to suboptimal local extrema. When the progress stalled, the algorithm either increases or decreases the LR depending on the behaviour of the MI levels beforehand.

This method makes optimization more responsive, making it less reactive to sudden changes and dissimilarities between experiments.

5 | Results

Patient ID	slice	MI with constant learning rate [bits]	MI with varying learning rate [bits]
1	1	1.07374033	1.07672728
1	2	1.034412	1.02637409
1	3	0.9673508	0.97274125
2	1	0.98804054	0.99761667
2	2	0.97438817	0.98468476
2	3	0.9468492	0.94289275
3	1	1.06241391	1.07077463
3	2	1.0228942	1.06736559
3	3	0.96821846	1.00484147
mean		1.0043	1.0160
std		0.0455	0.0476

Table 5.1: The best Mutual Information (MI) given for the registration between a T1 and T2 weighted MR image for each patient and slice with both methods.

The performance was evaluated by comparing the maximum Mutual Information (MI) that was reached by both the constant and varying learning rate strategies on the 9 slice pairs. As showed in Table 5.1, the learning rate strategy yielded a slightly higher average MI compared to the constant learning rate. (Respectively a mean of 1.0160 vs 1.0043). Statistical analysis revealed a non-significant trend towards higher MI for the varying learning rate (paired t-test: $p = 0.080$) assuming normal distribution according to normality test (Shapiro-Wilk: $p = 0.087$).

While the values of the MI were similar for both approaches, the convergence behaviour was different between the two methods. This can be seen in the plots in the Appendix: A. The plots of the constant learning rate show more aggressive oscillations throughout the optimization process, while the varying learning rate makes the oscillations less noticeable over time, due to the changing rate. However, despite these changes in the convergence behaviour, the differences in the final registration accuracy are not significant.

6 | Discussion

In this study, we implemented and compared a constant learning rate and a dynamic varying learning rate strategy for inter-modal registration, using MI as a similarity metric. Although the adaptive method was designed to make updates to prevent local optima and plateaus, our results indicate that both approaches yielded similar performance in terms of final registration accuracy and convergence iterations. The hypothesis is partially supported: adaptive LR achieved a marginally higher mean MI (1.016 vs 1.004), but the difference was not statistically significant ($p=0.080$). This suggests that the current implementation may be too conservative or not sufficiently sensitive to the gradients in the MI function. The used dataset may also not be complex enough to really profit from having an adaptive learning rate. To improve registration performance and convergence robustness, several extensions and adaptations to the current method can be considered:

6.1 | Design of the experiment

While the choice for mutual information instead of cross correlation is, for our case, the correct one, one limitation of the current setup is the exclusive use of standard MI. This can be sensitive to intensity scaling and in some circumstances non-comparable across scans. To create a more robust mechanism for more standardisation and to be able to compare it to literature, it's possible to calculate the mutual information is by calculating the entropy of both images and combining it into a normalised mutual information.

$$H(A) = \sum_j (\sum_k PDF[j, k] \log \sum_l PDF[j, l]) \quad \text{Entropy A} \quad H(B') = \sum_k \left(\sum_i PDF[i, k] \log \sum_j PDF[j, k] \right) \quad \text{Entropy B}$$

The joint entropy is calculated using the entire joint histogram. From these, the normalized mutual information (NMI) is derived as follows:

$$(NMI(A, B') = \frac{H(A) + H(B')}{H(A, B')} + 1)$$

The "+1" is included to avoid division by zero and stabilize the measure during optimization. While trying to implement normalized mutual information, we found that NMI appeared nearly constant across iterations and did not provide meaningful gradients for both approaches; thus, we reverted to the standard MI. Further investigation into NMI computation could allow for more robust similarity measurement and comparison with existing literature.

6.2 | Code improvements

To improve the registration and convergence of the code, we could have implemented a function that calculates the length of the gradient vector and adjusts the learning rate based on the length, because a larger gradient vector length implies that the optimization process is far from an optimum. So larger gradient vector means a larger learning rate.

As explained in the methods, the current varying learning rate implementation adjusts the step size by fixed percentages (e.g. $\pm 5\%$ or $\pm 10\%$). A more responsive alternative could involve exponential scaling based on proximity to a target MI value. A commented suggestion in the code hints at this idea:

```
182 # make update fully dynamic? = 0.0001 + 0.0003 * (1.0 - MI_current)
```

This formula would gradually decrease the learning rate as NMI approaches 1.0, assuming 1.0 represents perfect registration. Implementing such a schedule could improve convergence stability and reduce the risk of overshooting optimal solutions.

6.3 | Algorithm Performance and Convergence Behavior

Although the varying learning rate function was designed to adapt based on the improvement of MI, it did not result in a significant difference from the constant LR. The learning rate decreased slowly over time, but this did not help avoid suboptimal solutions or improve convergence speed. It's possible that the images and transformations used in these tasks were not complex enough to benefit from a more dynamic strategy. The gradient landscape may have been smooth enough that a constant step size was sufficient. Additionally, more trial-and-error with parameter tuning may change the outcome of the MI values.

7 | Contributions

Name	Contribution
Rens	Report + Run code
Steef	Report + Run code
Aukje	Report + Run code
Bodo	Report + Code

8 | **AI disclosure statement**

This project made use of Claude by Anthropic as a supporting tool during development. Specifically, Claude was used for debugging and correcting code errors, improving sentence structure and written clarity, and as a sparring partner for exploring and evaluating code ideas.

All final decisions, implementations, and conclusions remain the work of the authors. AI-generated suggestions were critically reviewed and if needed adapted before adoption.

9 | References

- [1] Murphy K, Viergever MA, Pluim JP, Klein S, Staring M. elastix: a toolbox for intensity-based medical image registration. *IEEE Trans Med Imaging*, Jan;29(1):196-205, 2010.
- [2] Viergever MA, Pluim JP, Maintz JB. Mutual-information-based registration of medical images: a survey. *IEEE Trans Med Imaging*, Aug;22(8):986-1004, 2003.
- [3] Wells W. M. III, Viola, P. Alignment by maximization of mutual information. *International Journal of Computer Vision*, 24, 137–154., 1997.

A | Appendix: Registered images

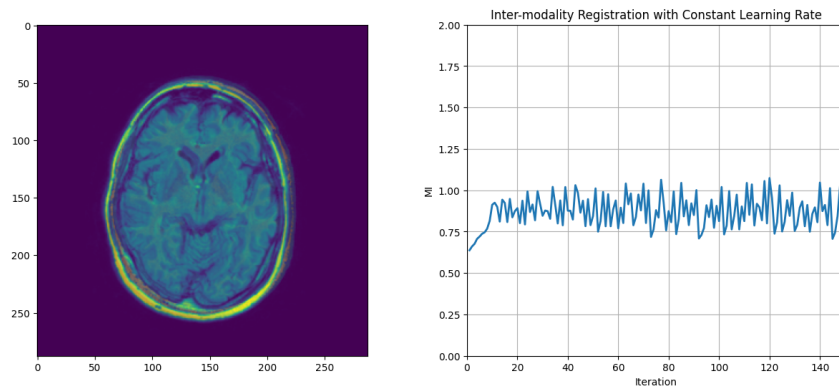


Figure A.1: Registered image with constant learning rate and MI per iteration for patient 1 slice 1

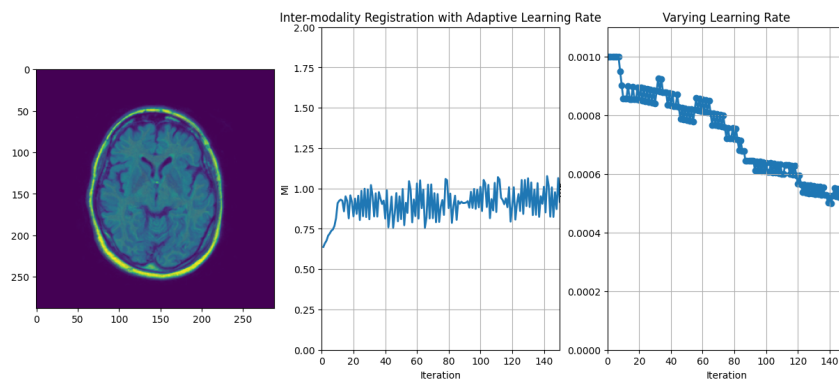


Figure A.2: Registered image with varying learning rate and MI per iteration for patient 1 slice 1

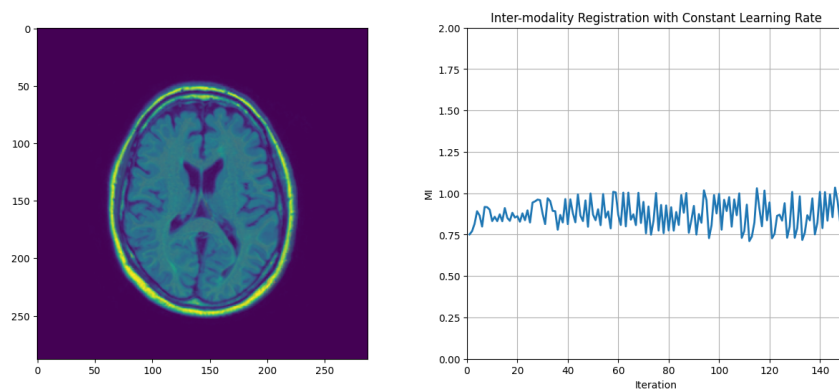


Figure A.3: Registered image with constant learning rate and MI per iteration for patient 1 slice 2

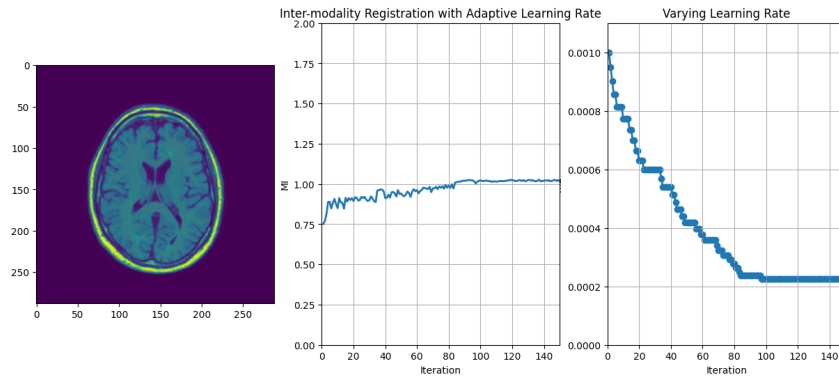


Figure A.4: Registered image with varying learning rate and MI per iteration for patient 1 slice 2

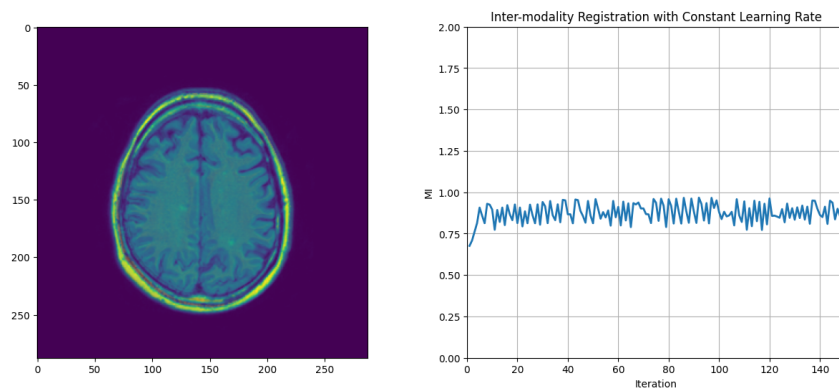


Figure A.5: Registered image with constant learning rate and MI per iteration for patient 1 slice 3

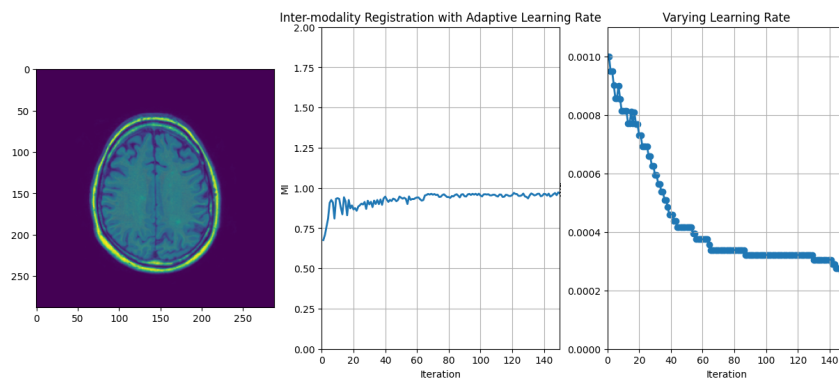


Figure A.6: Registered image with varying learning rate and MI per iteration for patient 1 slice 3

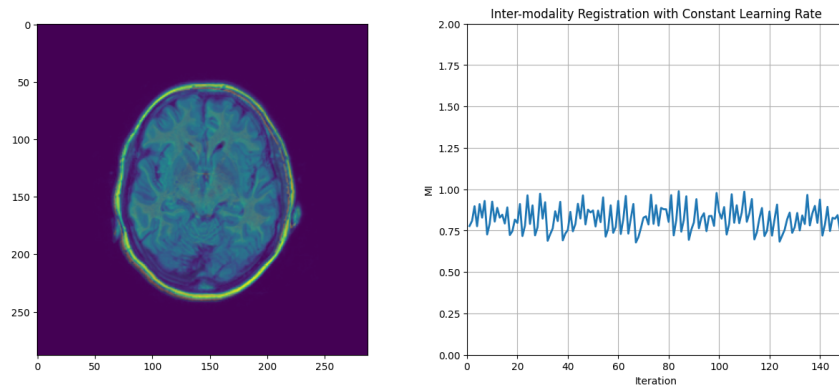


Figure A.7: Registered image with constant learning rate and MI per iteration for patient 2 slice 1

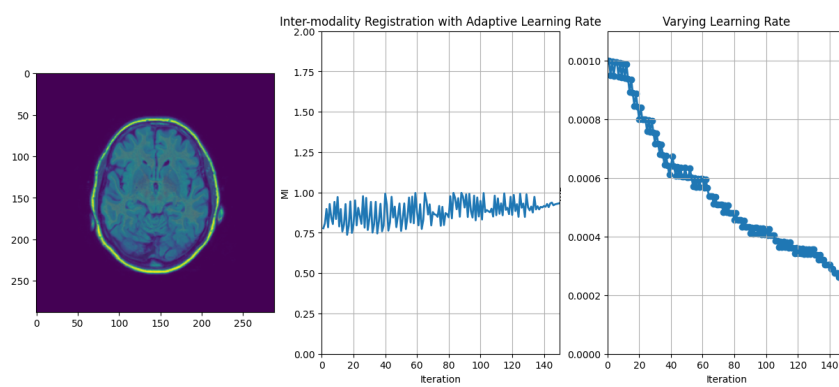


Figure A.8: Registered image with varying learning rate and MI per iteration for patient 2 slice 1

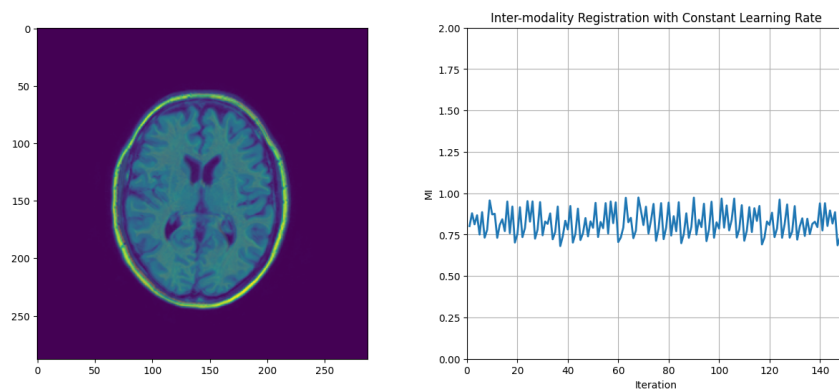


Figure A.9: Registered image with constant learning rate and MI per iteration for patient 2 slice 2

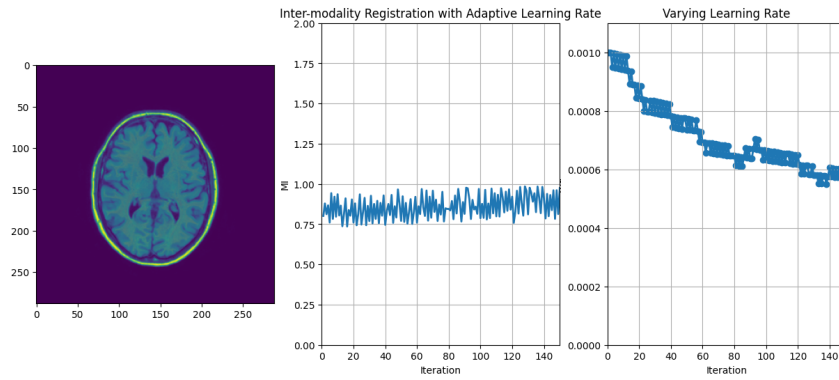


Figure A.10: Registered image with varying learning rate and MI per iteration for patient 2 slice 2

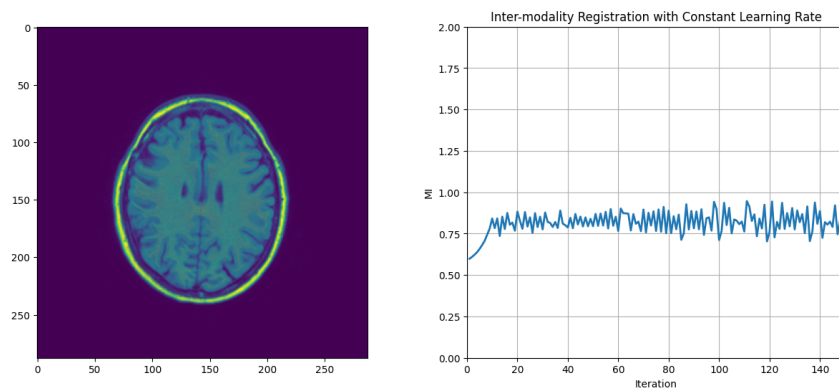


Figure A.11: Registered image with constant learning rate and MI per iteration for patient 2 slice 3

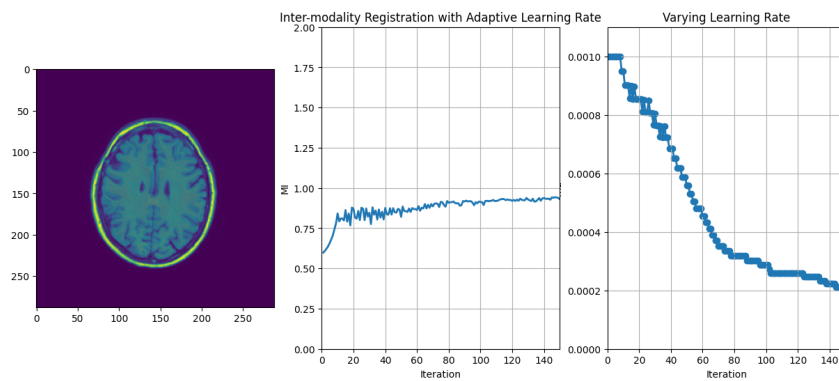


Figure A.12: Registered image with varying learning rate and MI per iteration for patient 2 slice 3

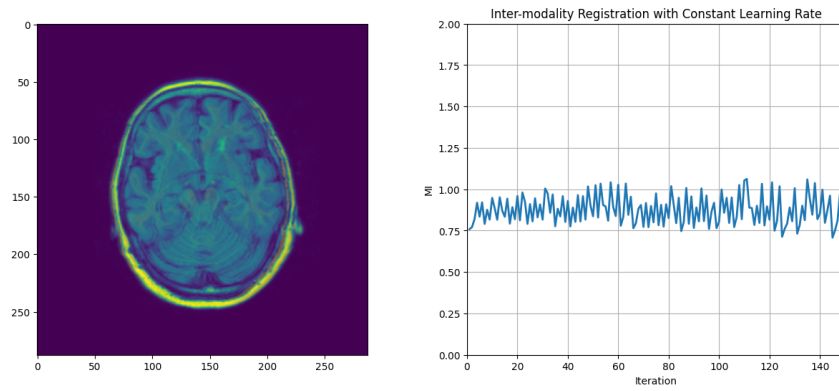


Figure A.13: Registered image with constant learning rate and MI per iteration for patient 3 slice 1

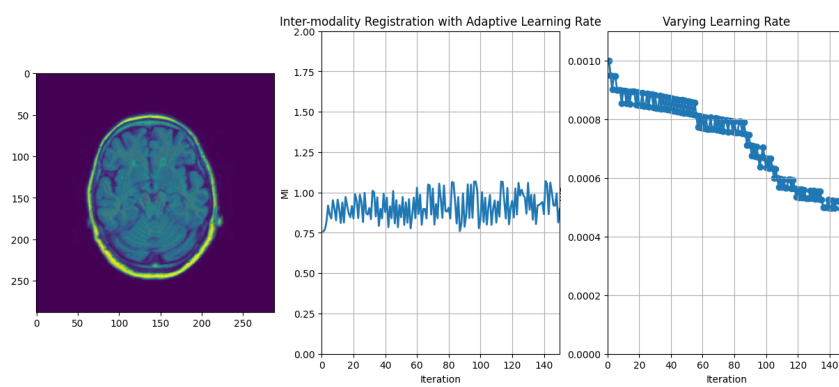


Figure A.14: Registered image with varying learning rate and MI per iteration for patient 3 slice 1

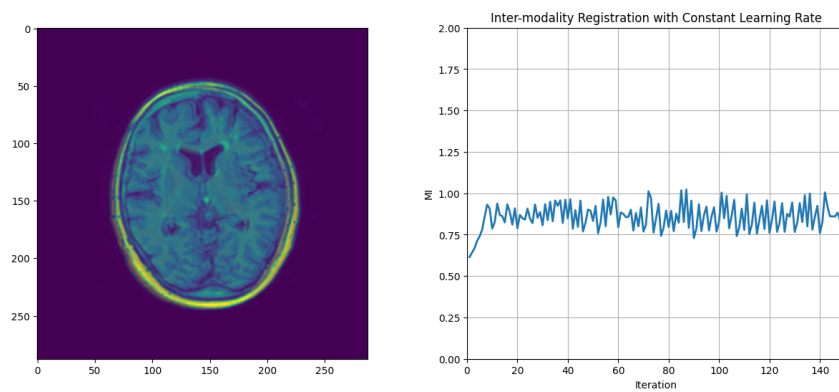


Figure A.15: Registered image with constant learning rate and MI per iteration for patient 3 slice 2

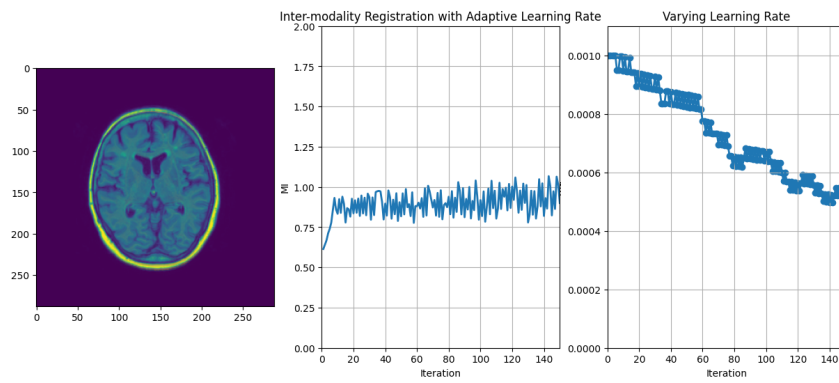


Figure A.16: Registered image with varying learning rate and MI per iteration for patient 3 slice 2

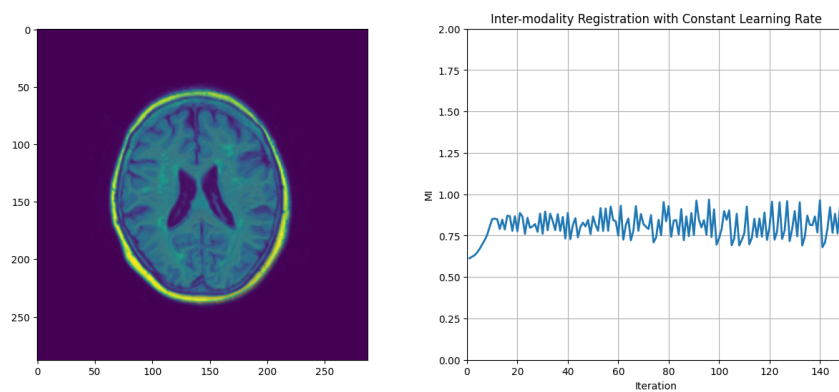


Figure A.17: Registered image with constant learning rate and MI per iteration for patient 3 slice 3

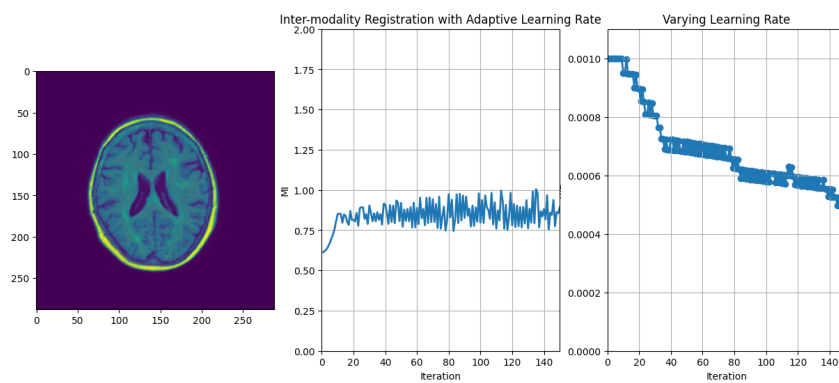


Figure A.18: Registered image with varying learning rate and MI per iteration for patient 3 slice 3